AT-ADD Track 2 Dataset Investigation Report

Project: Lightweight and Compression-Robust Audio Deepfake Detection for Edge Deployment

Purpose: Dataset logging, experiment planning, team reporting, and understanding of the dataset before downloading the full audio collection.

1. Dataset Overview

The investigated dataset is the AT-ADD Track 2 dataset. It is designed for all-type audio deepfake detection and contains four audio categories: speech, environmental sound, singing, and music. The metadata is currently available locally; the approximately 66 GB audio collection has not yet been downloaded. The analysis below is based on the three metadata files: t2_train_meta.csv, t2_dev_meta.csv, and t2_eval_meta.csv.

2. Metadata Structure

Each metadata file contains nine fields: name, type, label, source, source_id, speaker_id, generation_task, generator, and caption. The most important fields for our experiments are type (audio category), label (real/fake), generator (generation method), speaker_id (where available), source, and generation_task.

3. Dataset Size

Split	Total	Speech	Sound	Singing	Music	Fake	Real
Train	146,781	49,575	39,840	36,000	21,366	113,631	33,150
Dev	91,069	49,734	19,929	16,000	5,406	71,593	19,476
Eval	229,373	144,078	28,593	24,332	32,370	184,957	44,416

Total metadata records across the three splits: 467,223. The evaluation split is substantially larger than either training or development split.

4. Key Observation About speaker_id and generator

A very important metadata property was discovered in the speech subset. Real speech samples contain usable speaker_id values and have generator='-'. Fake speech samples have generator values but speaker_id='-'. This is not simply random missing data; it follows the real/fake construction of the metadata. Therefore, we must not discard samples merely because one of these fields is '-'. For fake speech, generator is the relevant identity for generalization; for real speech, speaker_id is useful for speaker-overlap analysis.

5. Speech Generator Analysis

Split	Fake samples	Fake generators	Generator relationship
Train	39,576	20	20 generators
Dev	39,734	20	Same 20 as train
Eval	124,078	25	25 completely unseen

Speech provides a particularly strong unseen-generator experiment: the 20 fake generators present in train are also present in development, while none of those 20 appears in evaluation. Evaluation instead contains 25 different generators. Thus the evaluation set can test whether a detector learned general characteristics of synthetic speech rather than memorizing the training generators.

6. Non-Speech Generator Analysis

Type	Train fake	Train gens	Eval fake	Eval gens	Train/Eval overlap
Sound	29,986	3	19,673	4	0
Singing	27,000	3	19,332	5	0

Music	17,069	4	21,874	4	0

Sound training fake generators are AudioLDM, AudioLDM2, and Audiogen; evaluation generators are Auffusion, Both-Ears-Wide-Open, MeanAudio, and UniAudio2. Singing training generators are NeuCoSVC_inhouse, SeedVC, and So-VITS-SVC_inhouse; evaluation generators are Free-SVC, KNN-SVC, RVC-FSD, So-VITS-SVC-FSD, and VEVO1.5. Music training generators are AudioLDM2, MusicGen, MusicLDM, and Stable Audio open; evaluation generators are ACE-Step, Musictango, SSM-TTM, and Uniaudio2. There is no train/evaluation generator overlap in any of the three non-speech categories.

7. Speaker Overlap Check for Real Speech

Comparison	Overlapping real speakers
Train ∩ Dev	8
Train ∩ Eval	144
Dev ∩ Eval	13

Real-speech speaker identities are largely separated across the official splits, but the separation is not perfect. Therefore, we should not describe the complete binary speech evaluation as perfectly speaker-disjoint. The strongest defensible claim is that generator-disjoint evaluation is available, while speaker overlap was explicitly inspected and found to be limited but non-zero.

8. Speech Sources and Generation Tasks

Speech real samples come from sources including LibriTTS-R, LJSpeech, Common Voice, AISHELL-3, and other sources; the evaluation real set additionally contains EchoFake and 3D-Speaker. Fake speech is associated with generation tasks including TTS, vocoder, voice cloning, and voice conversion. This diversity is useful because a model can be tested beyond a single synthesis mechanism.

9. Implications for Our BTP

• **Primary generalization experiment:** Use generator-disjoint evaluation to measure whether compact detectors generalize to unseen generation methods.

• **Audio-type analysis:** Evaluate speech, sound, singing, and music separately and also report an overall result.

• **Model consistency:** E1, E2, and the stronger reference model should use the same core data partitions and evaluation metrics so that model comparisons are meaningful.

• **Leakage control:** Generator identity must not be used as an input feature. It should be used only to construct/describe the evaluation split.

• **Speaker analysis:** For speech, retain the real-speaker metadata for overlap analysis, but do not claim perfect speaker-disjoint binary classification because fake speech lacks speaker IDs.

• **Compression robustness:** Track 2 metadata alone does not establish compression robustness. Compression/degradation experiments will need to be applied later to the audio.

• **Edge deployment:** Dataset scale alone does not determine the final model. The compact CNN/CRNN baselines should be evaluated for accuracy together with model size, memory, latency, and compression effects.

10. Current Dataset Status

Completed: authenticated access, metadata download, train/dev/eval inspection, audio-type distribution, class distribution, speech generator overlap analysis, real-speaker overlap analysis, speech source/generation-task inspection, and non-speech generator-disjointness analysis.

Not yet done: full audio download, exact training subset selection, preprocessing, feature extraction, E1/E2 training, SSL reference training, robustness transformations, and edge deployment benchmarking.

11. Recommended Next Step

Before downloading the approximately 66 GB audio collection, define the exact subset to be used for E1/E2/E3. The subset should preserve the four audio types, real/fake balance, generator diversity, and the unseen-generator evaluation structure. After that, download only the required audio files if the metadata/file

layout permits selective retrieval; otherwise plan the full download with sufficient storage headroom.

12. Short Team Summary

AT-ADD Track 2 is a strong fit for our baseline/generalization objective. It contains speech, sound, singing, and music with real/fake labels and substantial generator diversity. Most importantly, the evaluation generators are completely unseen relative to training for speech, sound, singing, and music. This gives us a clean way to test whether lightweight detectors generalize beyond the generators seen during training. The metadata also revealed an important limitation: speaker IDs are available for real speech but not fake speech, so full speaker-disjoint binary evaluation cannot be claimed from this metadata alone. Compression robustness and edge deployment remain separate experiments that we must add ourselves.